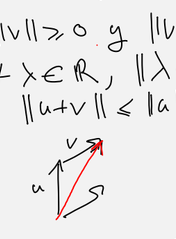
$N, \|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ :

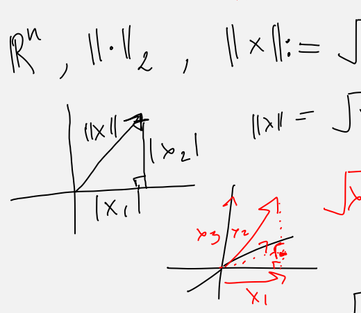
$\forall v \in \mathbb{R}^n, \|v\| \geq 0$ y $\|v\| = 0$ sii $v = 0$

$\forall v \in \mathbb{R}^n, \forall \lambda \in \mathbb{R}, \|\lambda \cdot v\| = |\lambda| \cdot \|v\|$

$\forall u, v \in \mathbb{R}^n, \|u+v\| \leq \|u\| + \|v\|$



$\mathbb{R}^n, \|\cdot\|_2, \|x\| := \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$



$\|x\| = \sqrt{x_1^2 + x_2^2}$

$\sqrt{x_1^2 + x_2^2 + x_3^2}$

$|\cdot|$ $\|\cdot\|_p : \|x\|_p = \sqrt[p]{\sum |x_i|^p}$

$= |\cdot|_1$ $\|\cdot\|_\infty : x = (x_1, \dots, x_n)$

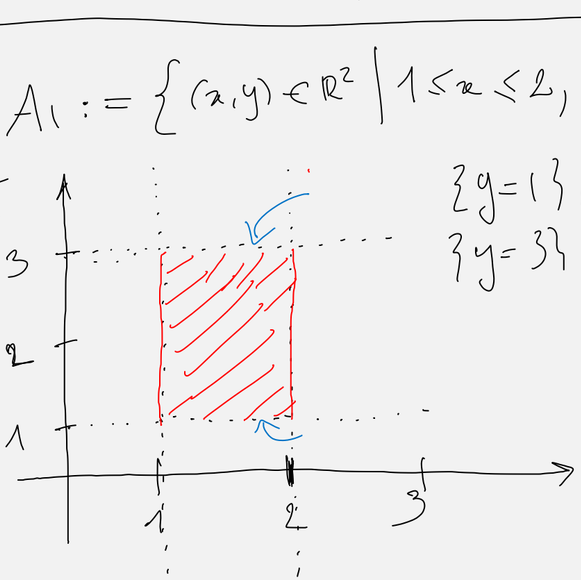
$\|x\|_\infty = \max\{|x_i|\}$

---

$x \in Adh(A)$ si: $\exists (x_n) \in A^{\mathbb{N}}$ tq $x_n \xrightarrow[n \to +\infty]{\|\cdot\|} x$ $\quad \overline{A}, \mathring{A}\ Int(A)$

$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N} \mid \forall n \geq n_0, \|x - x_n\| \leq \varepsilon$

$A_1 := \{(x,y) \in \mathbb{R}^2 \mid 1 \leq x \leq 2, 1 < y < 3\}$

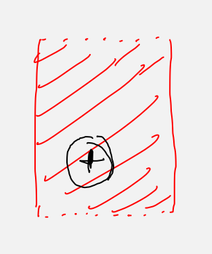


$\{y=1\} \cap A_1 = \emptyset$

$\{y=3\} \cap A_1 = \emptyset$

$B(x,r) := \{y \text{ tq } \|x-y\| < r\}$

$Int(A_1) = \{y \text{ tq } \exists \varepsilon > 0 \text{ tq } B(y, \varepsilon) \subset A\}$



$1 < x < 2 \quad \varepsilon > 0$

$1 < y < 3$

$B((x,y), \varepsilon) \subset A.$

$(a,b), \quad \|(x,y) - (a,b)\| \leq \varepsilon$

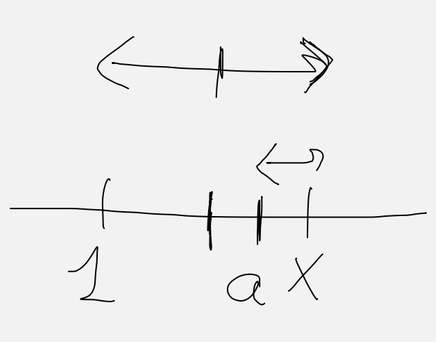
$\Leftrightarrow \sqrt{(x-a)^2 + (y-b)^2} \leq \varepsilon$

$\Leftrightarrow (x-a)^2 + (y-b)^2 \leq \varepsilon^2$

$\Rightarrow (x-a)^2 \leq \varepsilon^2, (y-b)^2 \leq \varepsilon^2$

$\Rightarrow |x-a| \leq \varepsilon, |y-b| \leq \varepsilon$

$\varepsilon < \min\left\{\begin{matrix} x-1, & 2-x \\ y-1, & 3-y \end{matrix}\right\} / 2$



$|x-a| \leq \varepsilon \Rightarrow a > 1$

$$\left.\begin{matrix} (a-x) = (a-1) + 1 - x \\ -\varepsilon \leq \quad \leq +\varepsilon \\ -\left(\frac{x-1}{2}\right) \leq \quad \leq \frac{x-1}{2} \end{matrix}\right\} \begin{matrix} (a-1) = (a-x) \\ \quad + (x-1) \\ \geq (x-1) - \frac{x-1}{2} > \left(\frac{x-1}{2}\right) \geq 0 \end{matrix}$$

Hacemos cuentas parecidas para mostrar: $\overset{a>1}{\phantom{x}}$ $a \underset{(\leq)}{<} 2,\ 1 < b < 3$

$\Rightarrow (a,b) \in A_1$ o sea: $B((x,y), \varepsilon) \subset A_1$

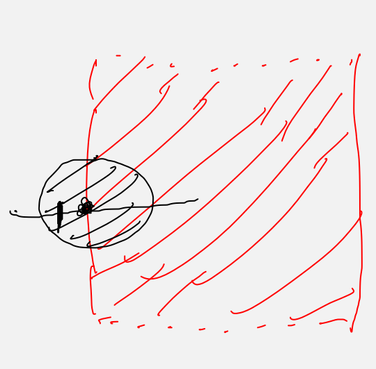
$\Rightarrow (x,y) \in Int(A_1).$

$p = (1,2) \overset{?}{\in} Int(A_1)$ ?

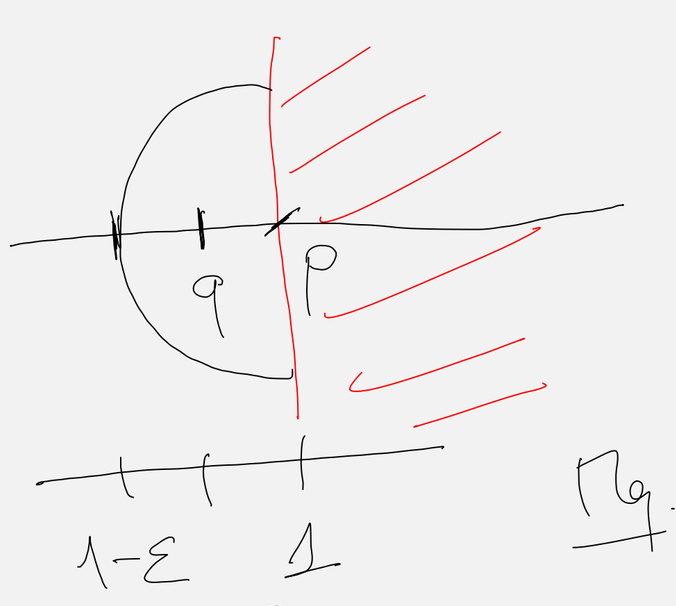
$1 \leq 1 \leq 2, \quad 1 < 2 < 3 \quad p \in A_1$

$p = (1,2).$ Pq $\forall \varepsilon > 0, B(p, \varepsilon) \not\subset A_1.$

$\Leftrightarrow \exists q \in B(p, \varepsilon)$ pero $q \notin A_1.$



$q = (1 - \frac{\varepsilon}{2}, 2)$



$q_x = 1 - \frac{\varepsilon}{2} < 1, \ q \notin A$

Pq $q \in B(p, \varepsilon)$

$\|p - q\| = \sqrt{(1 - 1 - \frac{\varepsilon}{2})^2 + \cancel{(2-2)^2}}$

$(\frac{\varepsilon}{2})^2$

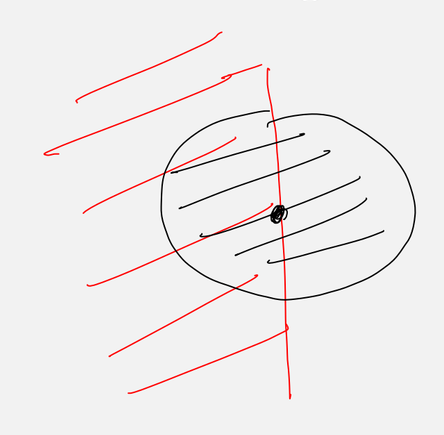
$= \sqrt{(\varepsilon/2)^2} = \varepsilon/2 < \varepsilon$

Si: $q \in B(p, \varepsilon).$

Entonces $p \notin Int(A).$

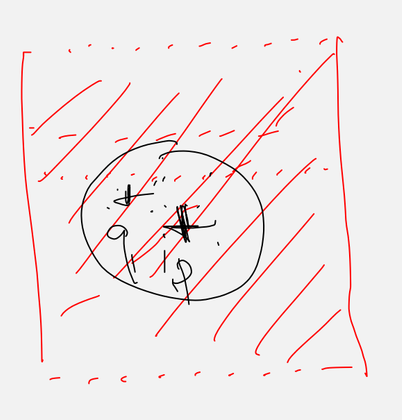
La misma prueba va a funcionar para

$p = (1, y) \quad \forall y \in (1,3)$



$p = (3, y)$

$B = \mathbb{Q} \cap A_1, \quad Int(B) = \emptyset, \quad Adh(A) = Fr(A) \sqcup Int(A)$



$p = (x,y) \in B \quad B(p, \varepsilon)$

$q = \underbrace{(x + \frac{\sqrt{2}}{n}}_{\notin \mathbb{Q}}, y + \frac{\sqrt{2}}{n})$

$\underbrace{Fr(A)}$ $\overset{\cup}{A}$ $\overset{\|}{A} \Rightarrow Int(A) = \emptyset$

$Int(A) = \left\{(x,y) : \begin{matrix} 1 < x < 2 \\ 1 < y < 3 \end{matrix}\right\}$

$A \subset Adh(A)$

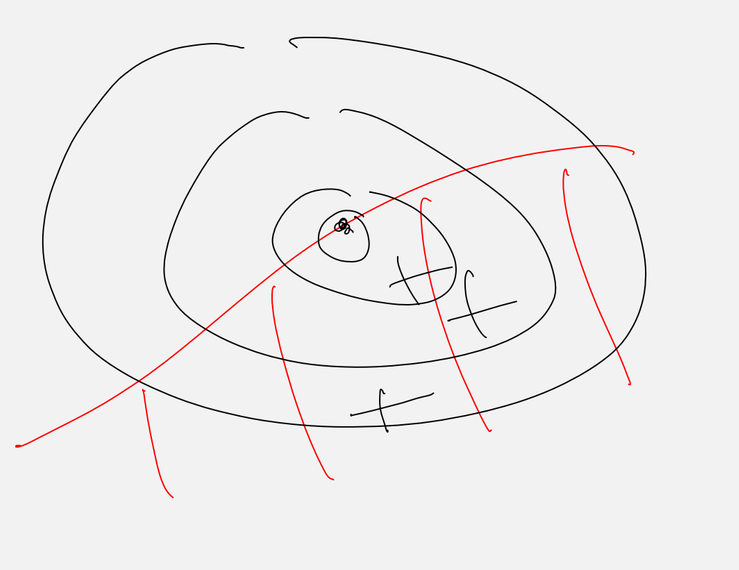
$a \in A, \quad a_n = a, \quad \underset{A}{\overset{\cap}{a_n}} \to a$

$a \notin A$ pero $a \in Adh(A).$

$\Leftrightarrow a$ tq $\forall \varepsilon > 0, B(a, \varepsilon) \cap A \neq \emptyset$

$\boxed{\Rightarrow}$

$B(a, 1/n) \cap A \neq \emptyset$



$\exists a_n \in B(a, 1/n)$

$\in A$

$a_n \quad \|a - a_n\| \leq 1/n \xrightarrow[n \to +\infty]{} 0$

$\forall \varepsilon > 0, \exists n_0 \mid n \geq n_0 \Rightarrow \|a - a_n\| \leq \varepsilon$

o sea $a_n \to a.$

$\boxed{\Leftarrow}$ $\quad B(a, \varepsilon), \quad a_n \to a$

para $n \geq n_0$, $\|a_n - a\| \leq \varepsilon$

o sea: $a_n \in B(a, \varepsilon)$

o sea $A \cap B(a, \varepsilon) \neq \emptyset$

$\forall \varepsilon > 0,$

$p = (\frac{3}{2}, 1)$

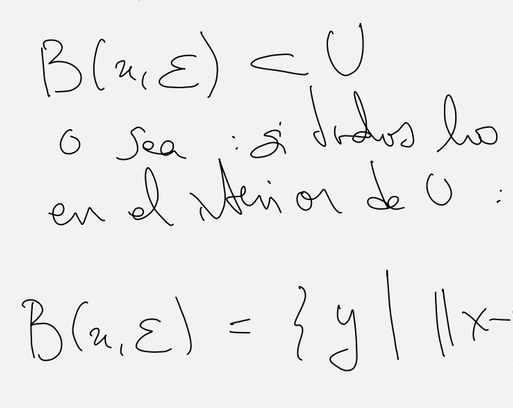
$B(p, \varepsilon)$

$(\frac{3}{2}, 1 + \frac{\sqrt{\varepsilon}}{2})$

$(\frac{3}{2}, 3 - \frac{\sqrt{\varepsilon}}{2})$

$$Adh(A) = \left\{(x,y) \;\middle|\; \begin{matrix} 1 \leq x \leq 2 \\ 1 \leq y \leq 3 \end{matrix}\right\}$$

$Fr(A) = Adh(A) \setminus Int(A)$



no A.

$Fr(A) \cap A \neq \emptyset$

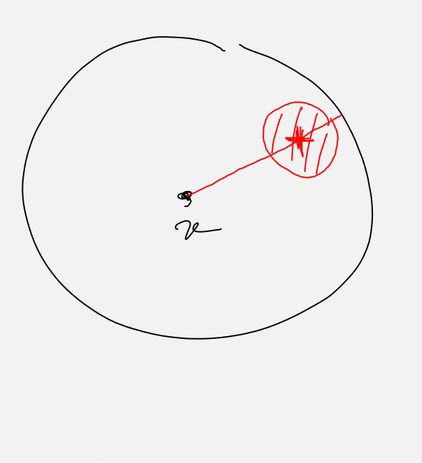
Abierto, Cerrado :

U es abierto si $\forall x \in U, \exists \varepsilon > 0$ tq

$B(x, \varepsilon) \subset U$

o sea : si todos los puntos de U están en el interior de U : $U = Int(U)$.

$B(x, \varepsilon) = \{y \mid \|x-y\| < \varepsilon\}$



no es abierto

no es cerrado